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Supplementary appendix

This appendix formed part of the original submission and has been peer reviewed. We post it as supplied by the authors.

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Supplementary appendix for: Pramipexole augmentation for the acute phase of Treatment-Resistant, Unipolar Depression: A UK-based, placebo-controlled, randomised trial.

Browning et al.

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Medical University of Vienna	Timea Helter
Trial Steering Committee membership	Rachel Upthegrove (chair), Michael Bradburn, Fiona Chan
Data Monitoring Committee membership	Bill Deakin (chair), Will Whiteley, Rebecca Woolley, Smitaa Patel

Full Inclusion and exclusion list

Inclusion Criteria:

At Screening:

1. Willing and able to give informed consent to participate in the trial
2. Males or females aged 18 years or over
3. Diagnosis of DSM-V major depression based on the affective disorder sections of the Mini International Neuropsychiatric Interview (MINI). N.b. comorbid anxiety disorder is not an exclusion criterion
4. Quick Inventory of Depressive Symptomatology self-report version (QIDS-SR16) score >10 (moderate, severe or very severe depression)
5. Currently taking an antidepressant medication
6. A lack of response to at least two antidepressants at therapeutic doses (based on Maudsley Prescribing Guidelines and/or British National Formulary) in the current episode
7. An indication for a change in treatment
8. Willing to continue antidepressant treatment
9. Women of child-bearing potential (WOCBP) only – a negative urine pregnancy test result

At randomisation:

1. Verbal consent to randomisation
2. QIDS-SR16 score >10
3. On a stable dose of an antidepressant for at least 4 weeks
4. WOCBP only – a negative urine pregnancy test result

Exclusion Criteria:

At Screening:

1. Clinical diagnosis of current or previous psychosis (including psychotic depression), bipolar disorder or Parkinson's Disease
2. Currently taking an antipsychotic medication
3. Clinically significant current or previous impulse control difficulties (determined during screening visit, included harmful alcohol or drug use)
4. Serious suicide or homicide risk
5. Current treatment with any medication known to interfere with pramipexole metabolism including cimetidine, memantine and methyldopa
6. Contraindications to pramipexole including history of or current treatment for eye disease (excluding near or long-sightedness), significant, symptomatic cardiovascular or renal disease or significant, symptomatic orthostatic hypotension
7. Previous course of pramipexole (>2 weeks)
8. Untreated or unstable medical condition which, in the judgement of the investigator, could interfere with the safety of receiving pramipexole or ability to complete the trial
9. Female and pregnant, breast-feeding or planning pregnancy
10. WOCBP not willing to use effective contraception

At randomisation:

1. Insufficient completion of questionnaires scheduled during the run-in phase (defined as at least 75% completion)
2. Has not responded to RA contact during the run-in phase
3. eGFR (from screening blood test) < 50 mL/min/1·73m2
4. Psychotherapy started in past 4 weeks or planned to start within next 12 weeks

Supplementary Methods

Statistical Analysis

Secondary outcomes

The continuous secondary outcomes were analysed in the same manner as the primary outcome. As the TSQM outcome is not collected at baseline (participants are unable to rate satisfaction with a treatment they have not yet received), analysis of this outcome did not include baseline score as an effect.

The binary secondary outcomes (response, remission, discontinuation of treatment due to intolerance) were analysed using a logistic mixed effects regression model with unstructured covariance structure. The model included randomised group, age, sex, and baseline QIDS-SR_{16} score as fixed effects, and site as a random effect. The adjusted marginal relative risk between the randomised groups and the corresponding 95% confidence interval were calculated from the model with delta-method standard errors.

The count secondary outcome (number of adverse reactions) was analysed using a Poisson mixed effects regression model with robust standard errors, and unstructured covariance structure. The model included randomised group, age, sex, and baseline QIDS-SR_{16} score as fixed effects, and site as a random effect. The adjusted incidence rate ratio between the randomised groups and 95% confidence intervals, and P-value were obtained from the model.

The assumption of the Poisson mixed effects regression model that the data follows a Poisson distribution was checked by plotting a histogram of the number of adverse reactions split by randomised group.

Additional safety outcomes, including SAEs, Adverse Events of Special Interest (AESI) and frequently occurring adverse events, were listed descriptively by group.

Analysis of health economic and mechanistic measures, including those used for mediator and moderator analysis, will be reported separately.

Sensitivity Analysis

Three pre-specified sensitivity analyses were used to examine the robustness of the results of the primary analysis.

Completeness of the primary outcome of change in the QIDS-SR_{16}, defined as completing the QIDS-SR_{16} questionnaire for at least 50% of the follow-up timepoints, was summarised by randomised group, and logistic regression models were used to explore any association between baseline characteristics and completeness of the QIDS-SR_{16}. The covariates found to predict the incompleteness were: QIDS-SR_{16} severity at baseline, and GAD-7 score at baseline. The primary analysis was reconducted with these factors included in the model (QIDS-SR_{16} score at baseline was not included as it was already in the model). The adjusted mean difference between the randomised groups and corresponding 95% confidence intervals were consistent with the primary analysis [-3·89 [-5·35 to -2·44], P<0·0001].

The second sensitivity analysis explored the impact of outliers. Any change in QIDS-SR_{16} from baseline at each follow-up time point that was more than ±3 standard deviations away from the mean was considered an outlier and set to missing. Four participants had a change

score that was considered an outlier (two at week 1, one at week 2, and one at week 6). All participants that had a change score that was considered an outlier were randomised to Pramipexole. No participants had a change score at the primary outcome timepoint of week 12 that was considered an outlier. The primary analysis was then reconducted with these outliers as missing. The adjusted mean difference between the randomised groups and corresponding 95% confidence intervals were again as for the primary analysis [-3·91 [-5·37 to -2·46], P<0·0001].

The normality assumptions of the linear mixed effects model were assessed by plotting a histogram of the primary outcome for each sensitivity analysis at each timepoint split by randomised group, and histogram of the model residuals, an inverse normal plot of the standardised model residuals, and a scatter plot of the fitted values versus the model residuals; these did not indicate substantial deviation from the model assumptions.

A pattern mixture model (PMM) was fitted to assess the robustness of the Missing At Random (MAR) assumption required for the linear mixed effect regression model. An additional PMM was conducted adjusting the multiple imputation model with the baseline covariates that were found to be predictive of the incompleteness of the QIDS-SR_{16} (GAD-7 score at baseline). These analyses indicated that, even across a range of missing data patterns, treatment difference would still be significant, indicating the robustness of the main results even with strong departures from the MAR assumption.

In summary, we explored the effect of baseline factors on missing data, the effect of excluding outliers and the effects of violations in the missing at random assumptions of the mixed effects model on the primary outcome result at 12 weeks in pre-specified sensitivity analyses. None of these factors was found to alter the reported effects of treatment on the primary outcome.

Supplementary Results

Secondary Outcomes

Efficacy

Figures S1-S5 present summary graphs of the secondary efficacy and acceptability outcomes. Summary statistics for all primary and secondary efficacy and acceptability outcomes are included in Table S2 at the end of this document.

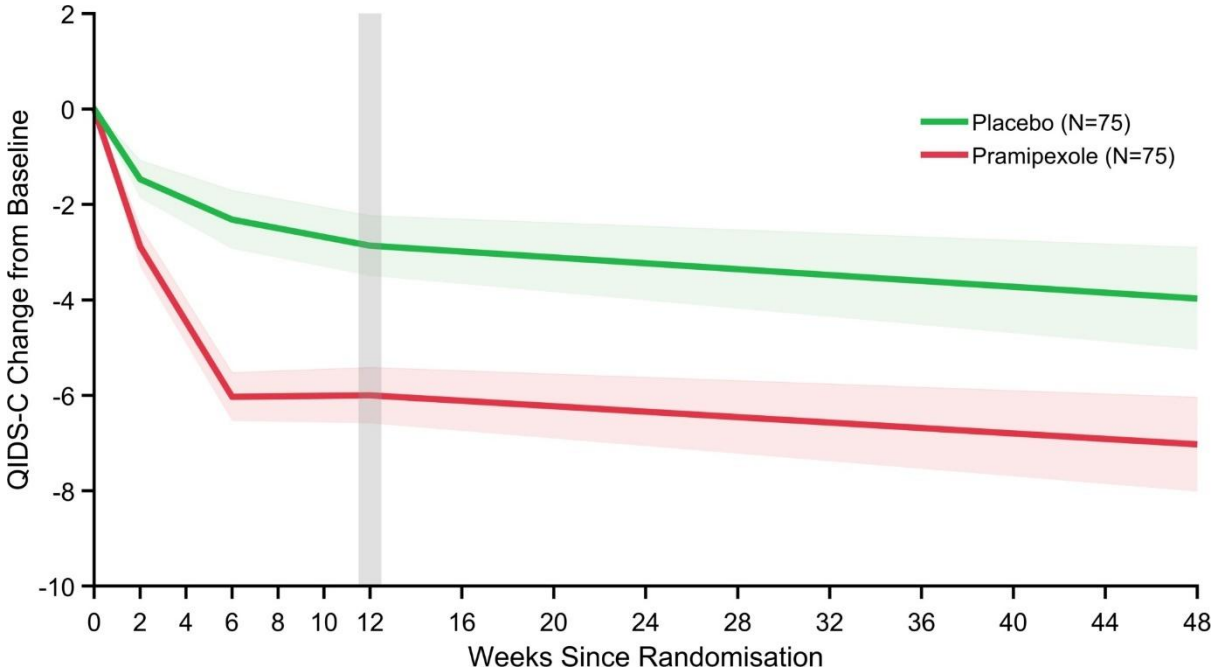

Figure S1: Change from baseline of clinician reported depression symptoms (QIDS-C). Lines report mean (SEM) of participant scores. Shaded grey region is timepoint of primary outcome. See main paper for results of prespecified statistical tests.

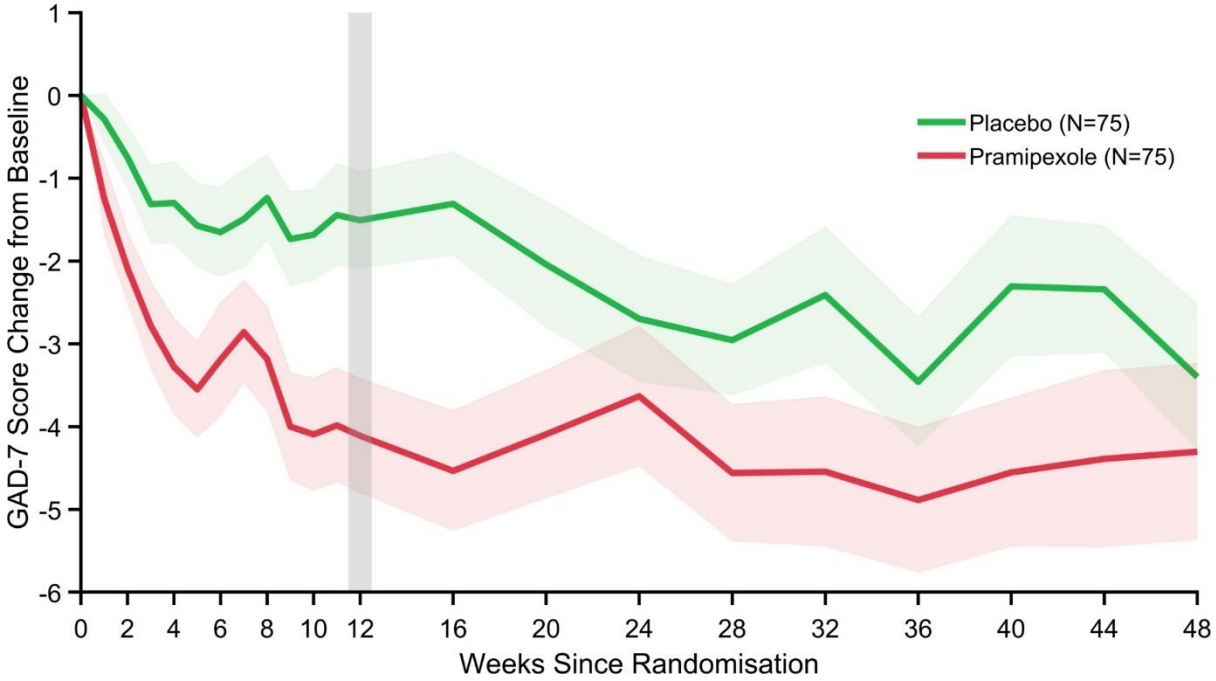

Figure S2: Change from baseline of anxiety symptoms (GAD-7). Lines report mean (SEM) of participant scores. Shaded grey region is timepoint of primary outcome. See main paper for results of prespecified statistical tests.

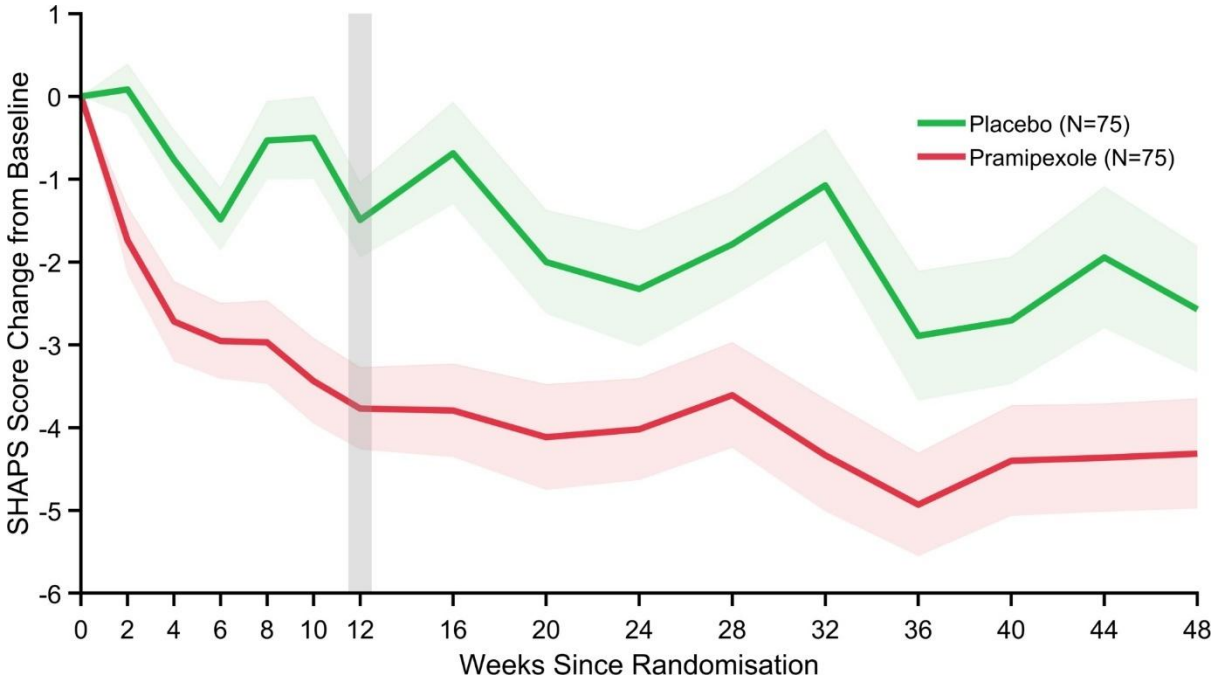

Figure S3: Change from baseline of anhedonia symptoms (SHAPS). Lines report mean (SEM) of participant scores. Shaded grey region is timepoint of primary outcome. See main paper for results of prespecified statistical tests.

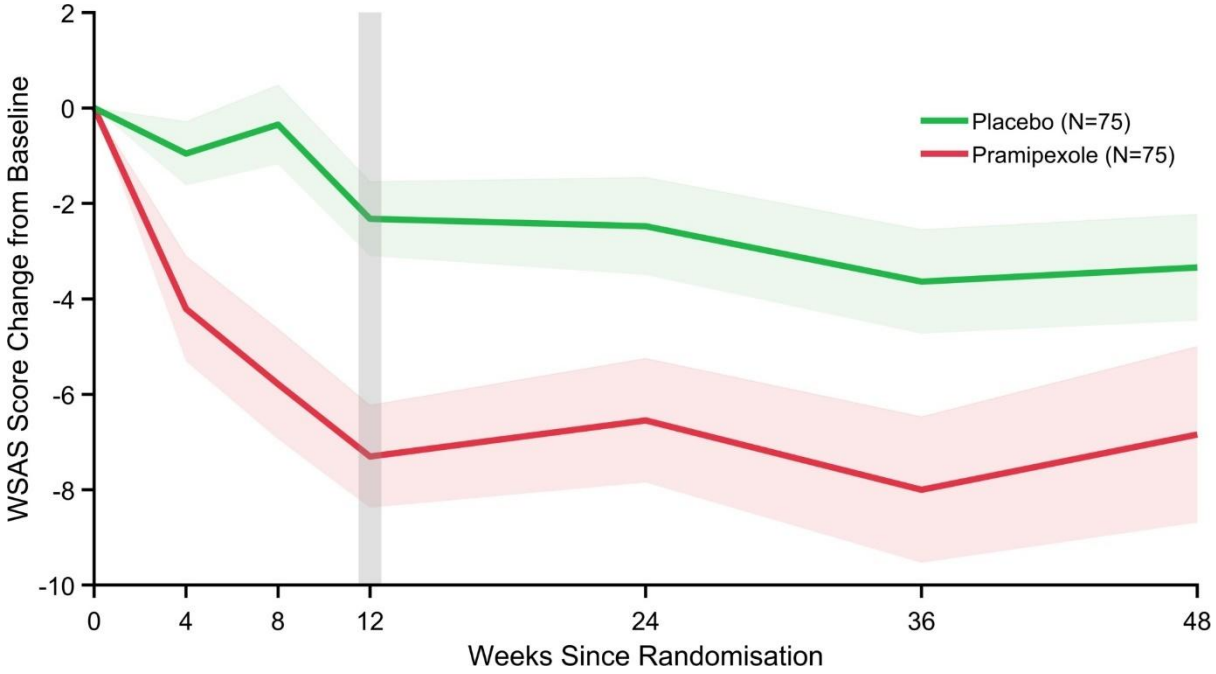

Figure S4: Change from baseline of functional outcome (WSAS). Lines report mean (SEM) of participant scores. Shaded grey region is timepoint of primary outcome. See main paper for results of prespecified statistical tests. Higher scores indicate greater functional impairment.

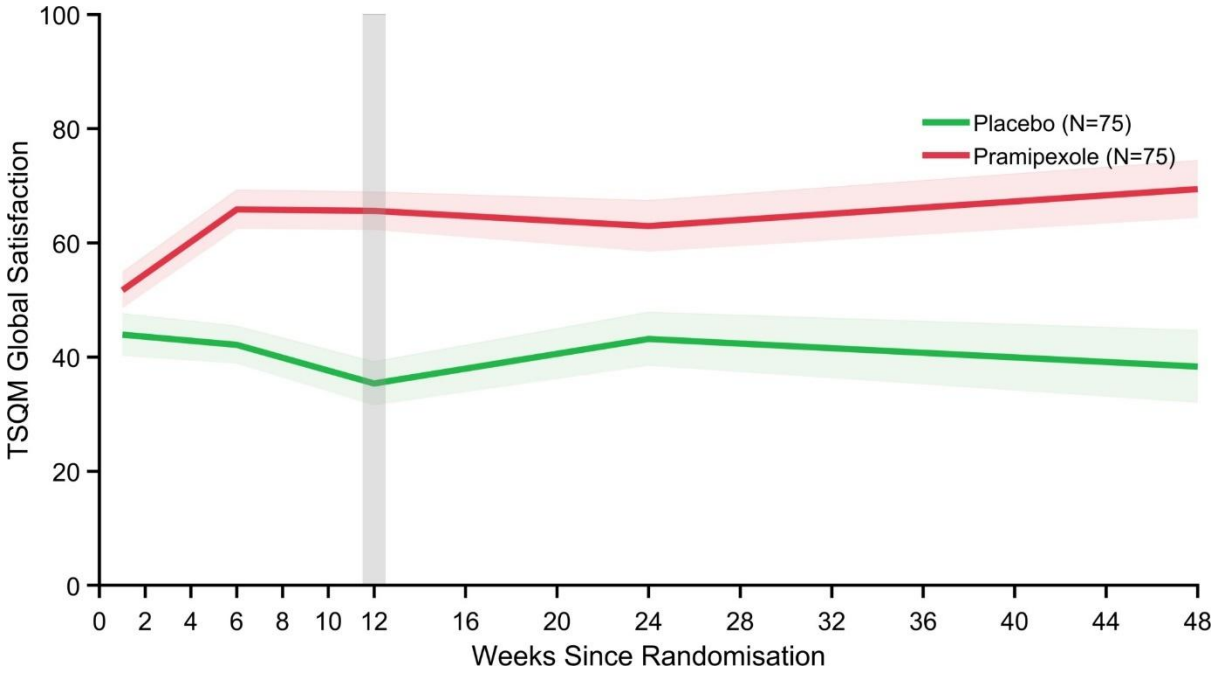

Figure S5: Mean score for global satisfaction with treatment (TSQM). Lines report mean (SEM) of participant scores. Shaded grey region is timepoint of primary outcome. See main paper for results of prespecified statistical tests. Higher scores indicate greater satisfaction with treatment. See Table S2 for other TSQM subscales.

Safety and tolerability

Figures S6-S8 present summary graphs of additional safety and tolerability outcomes. Summary statistics for all safety outcomes are included in Table S2 at the end of this document.

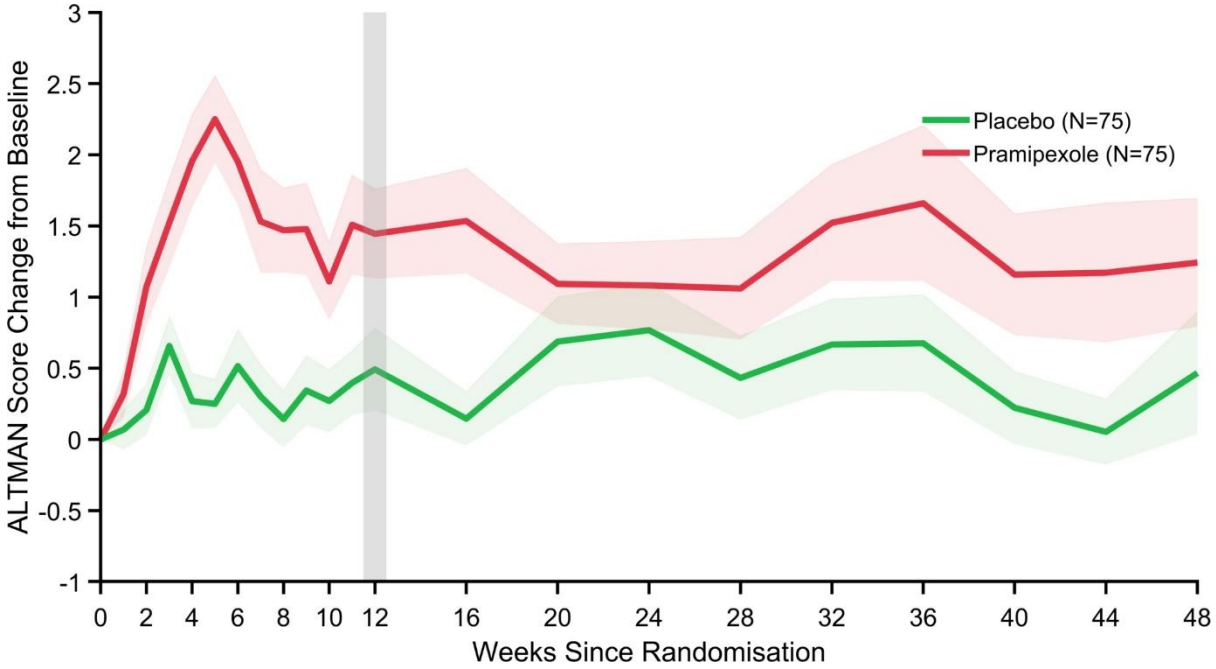

Figure S6: Change from baseline of mania symptoms (Altman). Lines report mean (SEM) of participant scores. Shaded grey region is timepoint of primary outcome. Higher scores indicate higher levels of symptoms of mania.

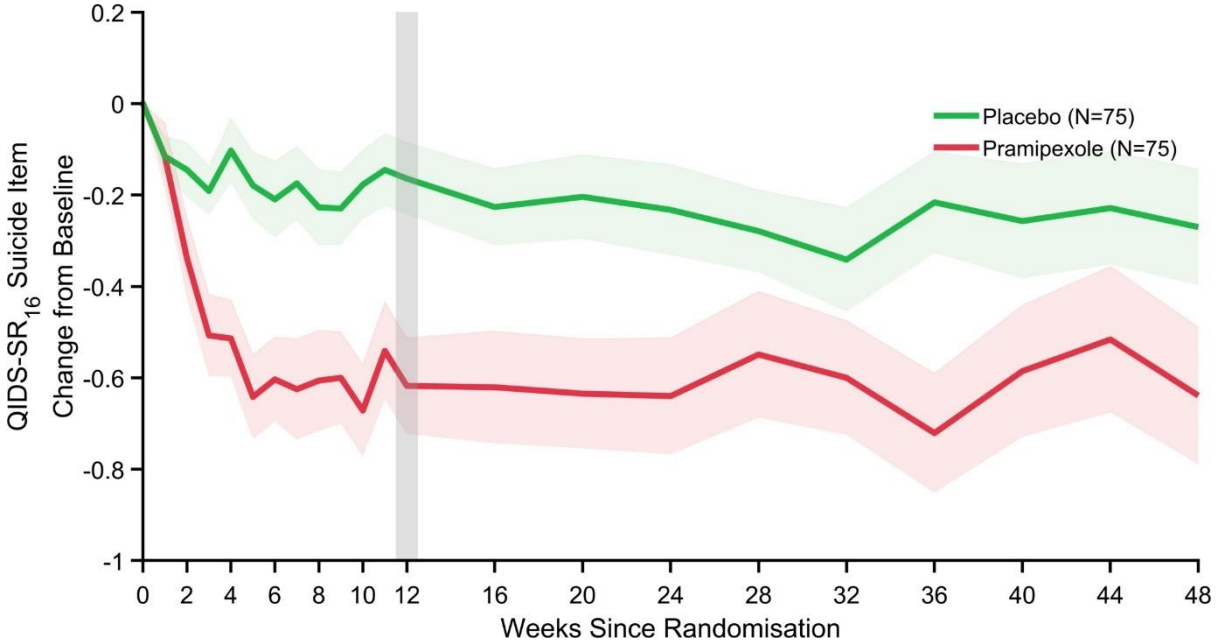

Figure S7: Change from baseline of suicide symptoms (Item 12 from QIDS-SR$_{16}$). Lines report mean (SEM) of participant scores. Shaded grey region is timepoint of primary outcome. Higher scores indicate a greater degree of suicidality.

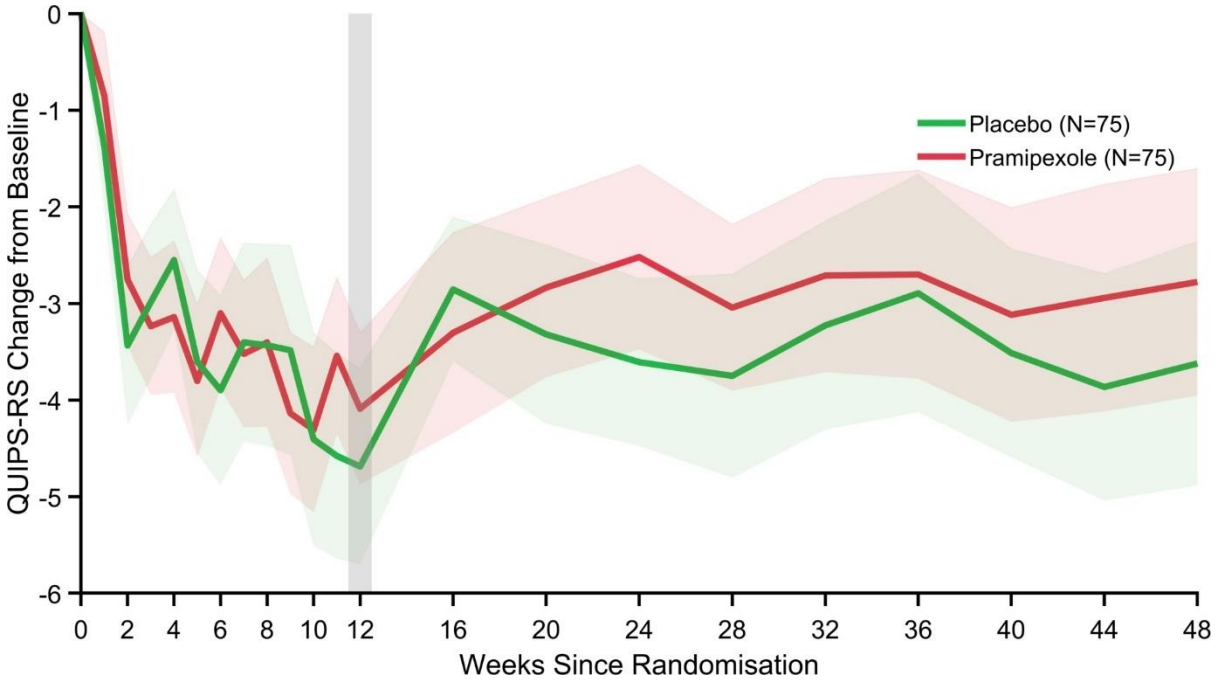

Figure S8: Change from baseline of impulse control difficulty symptoms (QUIPS-RS). Lines report mean (SEM) of participant scores. Shaded grey region is timepoint of primary outcome. Higher scores indicate more impulse control difficulties.

Table S1: Frequency and MEDRA codes for SAEs

	Pramipexole	**Placebo**	**Reason classified as SAE**
Preferred Term, *number of events*	**(N=75)**	**(N=75)**	
Strangulated hernia	1	0	Requires inpatient hospitalisation
Abscess	1	0	Requires inpatient hospitalisation
Arthritis infective	0	1	Requires inpatient hospitalisation
Post procedural infection	1	0	Requires inpatient hospitalisation
Ovarian cyst	1	0	Requires inpatient hospitalisation
Breast conserving surgery	0	1	Requires inpatient hospitalisation

Frequency of changes to antidepressant medication during the trial

Changes to concomitant antidepressant medication were not allowed in the first 12 weeks of the study (unless clinically mandatory). However, following this time point changes were allowed, and all changes to medication were regularly assessed (weekly until week 12 and then monthly until the end of the study). In total 18 (24%) participants in the pramipexole group and 36 (48%) participants in the placebo group changed their antidepressant medication at some time during the study. Figure S9 presents a histogram of the timepoints that change in antidepressant medication occurred (note a given participant could make more than one change). Participants in the placebo group changed their antidepressant medication across the course of the trial at a higher rate than participants who received pramipexole (mean (SD) changes of antidepressants across the trial in pramipexole group 0·57 (1·23), placebo group 1·77 (2·54)). As can be seen in Figure S9, the frequency of change of antidepressant medication in the placebo group was greater than that in the pramipexole group at all timepoints.

A total of 6 participants in the pramipexole group and 19 participants in the placebo group made at least one change to their antidepressant treatment before the week 12 primary outcome. These changes were discouraged in the protocol, although were allowed to occur if deemed clinically necessary by the treating clinician. To the extent that these changes influenced subsequent depression rating scores, the higher rate of change in the placebo group would be expected to reduce the observed difference between the pramipexole and placebo groups.

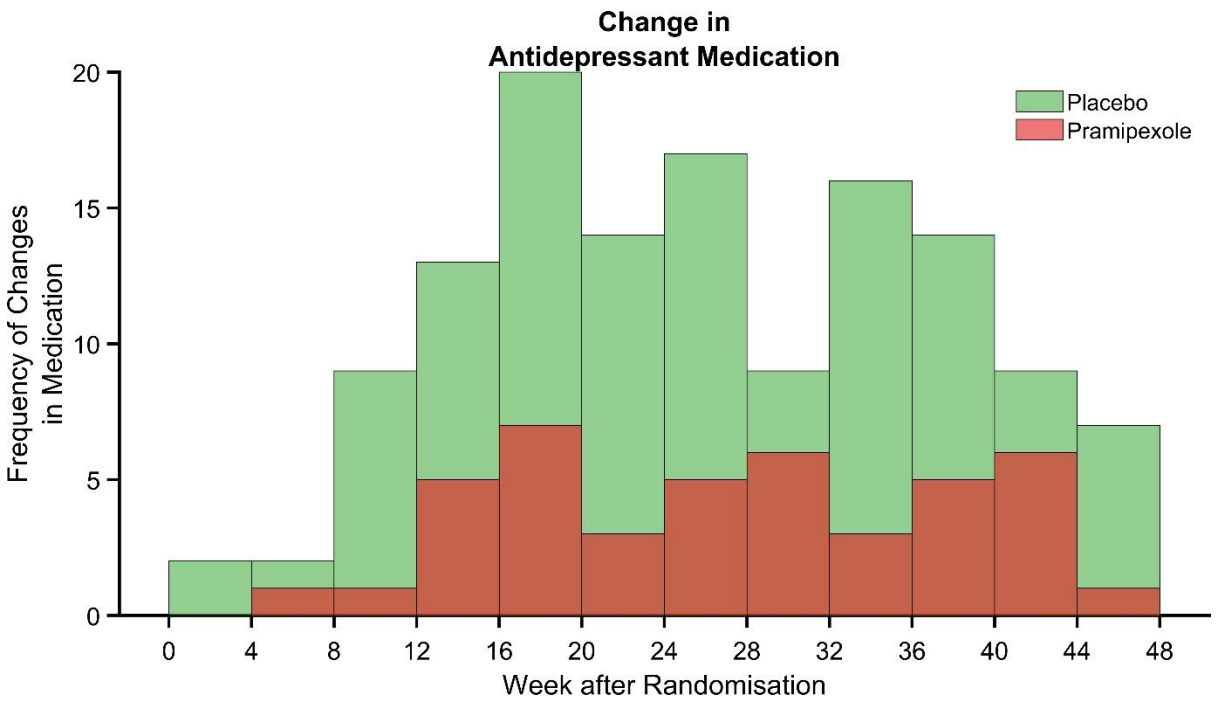

Figure S9: Histogram of the frequency of changes in antidepressant medication in four week time bins across the study. As can be seen, the placebo group changed their antidepressant medication more frequently than the pramipexole group across the trial.

Functional unblinding

Perceived group allocation was collected from 128 participants at the end of study involvement (63 in pramipexole group, 65 in placebo group). Participants were asked to guess whether they were taking pramipexole or placebo and then to rate their confidence in this guess (0= not at all confident, 4= entirely confident). As can be seen in Figure S10 participants were able to guess their allocated group better than chance (exploratory analysis: Chi-Square test, $P<0{\cdot}001$), with confidence (for both correct and incorrect guesses) generally higher for participants who were receiving placebo.

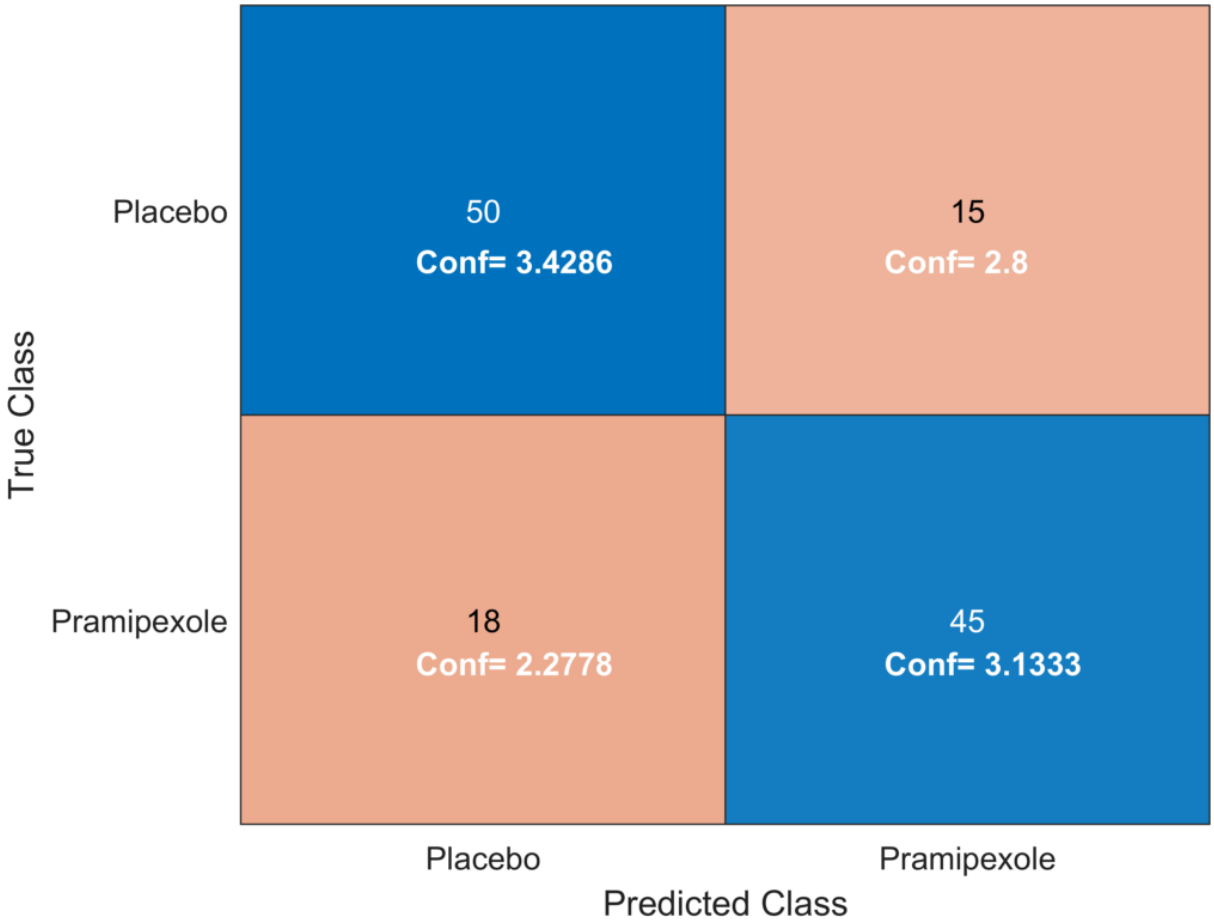

Figure S10: Perceived allocation of participants at the end of the study. The true class describes the group participants were allocated to (i.e. whether they actually received pramipexole or placebo), the predicted class is the perceived group (i.e. whether they thought they were receiving pramipexole or placebo). The numbers at the centre of each square are the number of participants in each cell, with the mean confidence score for the guess reported underneath (from 0=not at all confident, 4=entirely confident).

Figure S11 reports the mean (SEM) of the primary outcome (change in QIDS-SR_{16} from baseline to week 12) in participants who were matched in terms of perception of treatment. Specifically, those who a) thought they were receiving placebo (left hand columns), and b) thought they were receiving pramipexole (right hand columns). In both cases, participants

who were actually taking pramipexole reported a numerically greater decrease in symptoms of depression than those taking placebo.

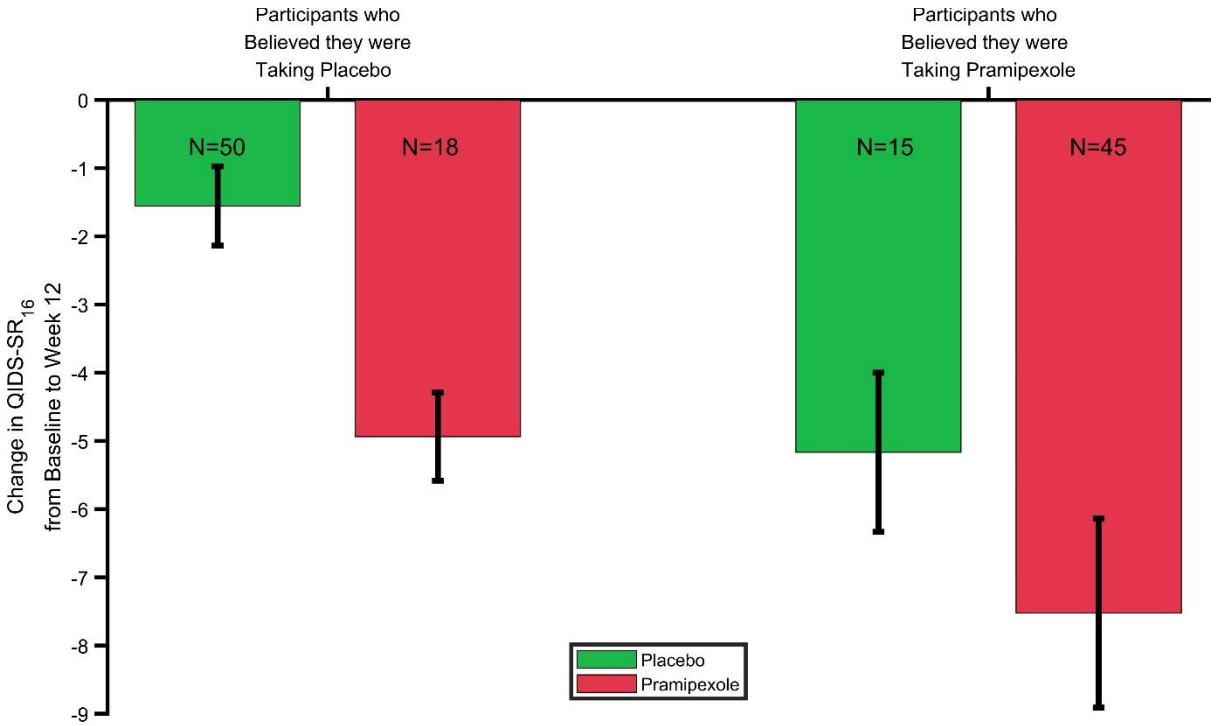

Figure S11: The primary outcome of the study (change in QIDS-SR16 from baseline to week 12) in participants split by perceived treatment (left side: Participants who guessed they were on placebo, right side: Participants who guessed they were on pramipexole) and actual treatment (pramipexole= red, placebo = green). Bars report the mean (SEM) change in QIDS-SR$_{16}$ scores.

Exploratory subgroup analysis

As a dopaminergic agent, pramipexole might be expected to be particularly active against symptoms of anhedonia. As reported in the main paper, there was clear evidence that pramipexole treatment reduced symptoms of anhedonia, measured using the SHAPS. This raises the possibility that pramipexole is particularly effective in participants who have high levels of anhedonia at baseline. This was tested in an exploratory subgroup analysis by including baseline SHAPS scores in the statistical model of the primary outcome and assessing for an interaction effect between the SHAPS and treatment response at week 12. This interaction effect was not significant P=0·74.

Non pre-specified analyses

The reported analyses and figures below were not pre-specified in the statistical analysis plan.

Primary outcome per site

Figure S12 reports on the primary outcome (change in the QIDS-SR_{16} from randomisation to week 12) split by study site. As reported in Table 1 of the main paper a number of sites had very few participants. Therefore, data are displayed individually for sites which randomised at least 5 participants per group, with data for the remaining sites collected into the "others" category. The number of randomised participants for each site and group are displayed as numbers above each bar. As can be seen, the relative benefit of treatment is reasonably consistent across sites.

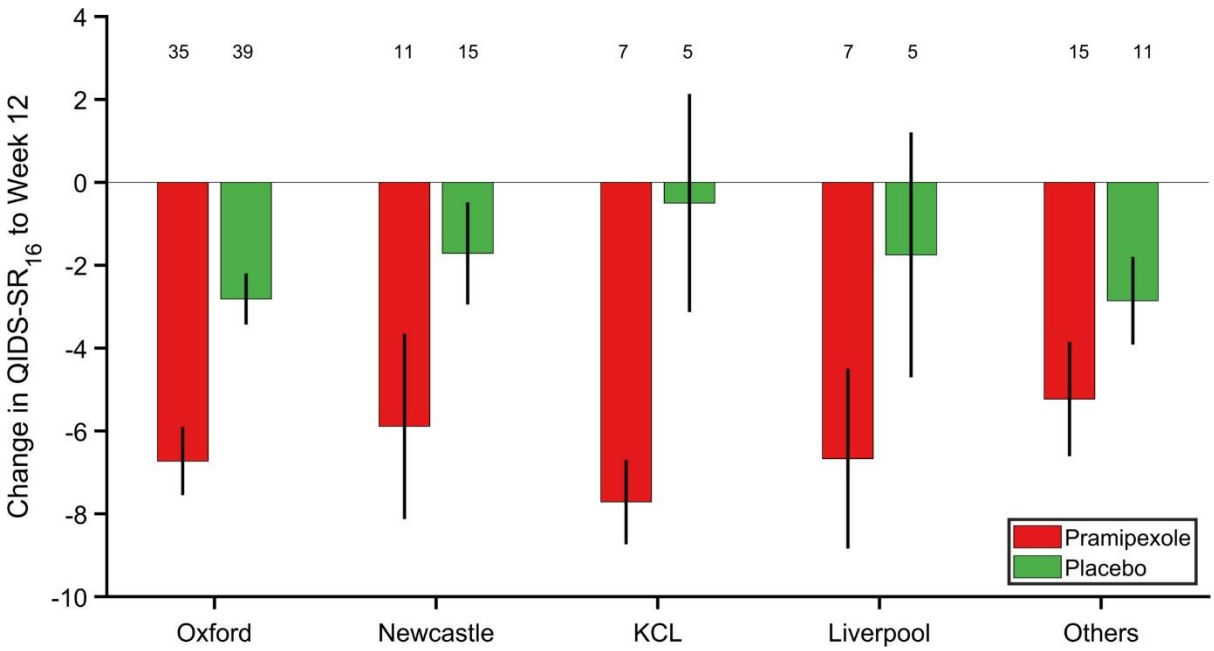

Figure S12: The primary outcome of the study split by study site. The graph reports the change from randomisation to week 12 of the QIDS-SR_{16} split by study site. The Oxford, Newcastle, KCL (King's College London) and Liverpool sites are displayed separately as these sites randomised at least 5 participants to each treatment arm. The other sites (Bristol, University College London, Pennine, Midlands, Nottingham), that recruited fewer participants, are combined into the "others" category. Bars display the mean change in scores, error bars report the standard error of the mean, the numbers above each bar are the number of randomised participants per group in each site.

Causes and frequency of discontinuation of study drug

Participants who discontinued study medication were asked why they had made this decision. Discontinuation of study drug due to intolerance (which was defined as people stopping the IMP or withdrawing from the study and giving the reason for this dropout as an adverse event or serious adverse event) was a prespecified outcome in the trial. As reported in the main manuscript, this occurred more frequently in the pramipexole than the control group. This raises the possibility of differential attrition between the two groups. However, side effects were not the only reason participants stopped the study drug, they also discontinued because of a perceived lack of benefit, or because they were unable to comply with study procedures or simply withdrew consent without giving any reason for this decision. Figure S13a summarises data on the reported reasons for discontinuation. As can

be seen while participants on pramipexole tended to stop the drug because of side effects, participants in the placebo group tended to stop because of lack of perceived benefit. Figure S13b illustrates the proportion of randomised participants who remained on the study drug across time. As can be seen, the rate of discontinuation was well matched between groups.

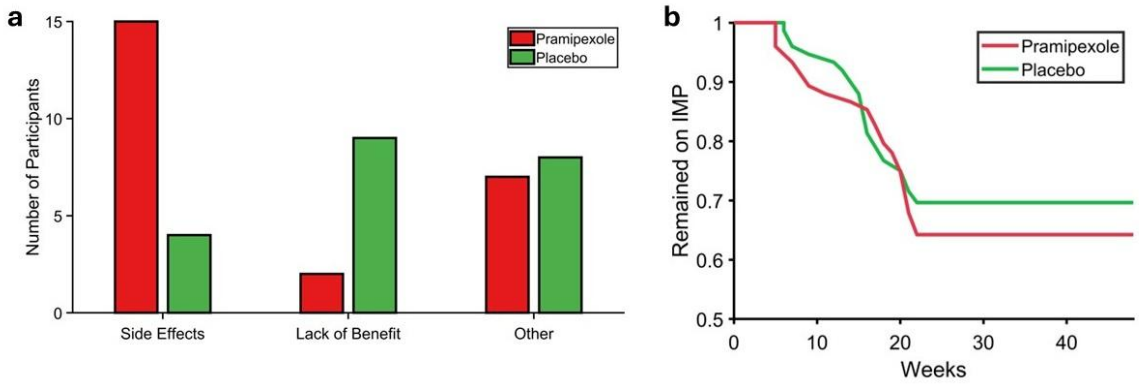

Figure S13: ***a*** *Frequencies of the reported reasons that participants discontinued study medication, split by treatment group. The most common reasons for discontinuing were side effects (more common in those randomised to pramipexole) and perceived lack of benefit (more common in those randomised to placebo). The "other" group includes participants who did not give a reason for withdrawal, or who could not comply with study procedures, and was similar between groups.* ***b*** *The proportion of randomised participants in each group who remained on the study drug at each time point. Displayed data was censored at study closure (i.e. participants who did not complete the study because it closed before they reached week 48 are not counted as discontinuing the medication).*

Table S2: Summary statistics, adjusted treatment effects, and P-values for all primary and secondary efficacy, acceptability and safety analyses

	Pramipexole (N=75)	Placebo (N=75)	Treatment effect [95% CI]*	P-value†
Quick inventory of depressive symptomatology (QIDS-SR$_{16}$)‡				
QIDS-SR$_{16}$ change at week 12 (Primary outcome)				
Adjusted mean difference			-3·91 [-5·37 to -2·45]	<0·0001
Mean (SD)	-6·4 (4·9)	-2·4 (4·0)		
[Range]	[-19 to 4]	[-13 to 4]		
Missing	*7*	*8*		
QIDS-SR$_{16}$ change at week 1				
Adjusted mean difference			-0·91 [-2·36 to 0·53]	0·22
Mean (SD)	-1·8 (3·0)	-1·0 (2·6)		
[Range]	[-10 to 5]	[-7 to 7]		
Missing	*6*	*6*		
QIDS-SR$_{16}$ change at week 2				
Adjusted mean difference			-0·48 [-1·93 to 0·97]	0·51
Mean (SD)	-2·2 (3·8)	-1·8 (3·1)		
[Range]	[-15 to 7]	[-11 to 5]		
Missing	*7*	*6*		
QIDS-SR$_{16}$ change at week 3				
Adjusted mean difference			-1·87 [-3·32 to -0·42]	0·012
Mean (SD)	-4·0 (3·7)	-1·8 (3·2)		
[Range]	[-13 to 3]	[-8 to 7]		
Missing	*6*	*7*		
QIDS-SR$_{16}$ change at week 4				
Adjusted mean difference			-2·58 [-4·02 to -1·13]	0·00047
Mean (SD)	-4·6 (4·0)	-2·0 (3·4)		
[Range]	[-15 to 3]	[-10 to 8]		
Missing	*3*	*7*		
QIDS-SR$_{16}$ change at week 5				
Adjusted mean difference			-2·80 [-4·26 to -1·34]	0·00017
Mean (SD)	-5·3 (4·5)	-2·4 (3·8)		
[Range]	[-15 to 5]	[-12 to 6]		
Missing	*8*	*8*		
QIDS-SR$_{16}$ change at week 6				
Adjusted mean difference			-3·16 [-4·65 to -1·66]	<0·0001
Mean (SD)	-5·6 (4·7)	-2·6 (3·4)		

	Pramipexole (N=75)	Placebo (N=75)	Treatment effect [95% CI]*	P-value†
[Range]	[-18 to 4]	[-11 to 3]		
Missing	*17*	*13*		
QIDS-SR$_{16}$ change at week 7				
Adjusted mean difference			-2·69 [-4·16 to -1·21]	0·00037
Mean (SD)	-5·2 (4·6)	-2·4 (4·2)		
[Range]	[-15 to 4]	[-13 to 7]		
Missing	*11*	*12*		
QIDS-SR$_{16}$ change at week 8				
Adjusted mean difference			-2·95 [-4·41 to -1·48]	<0·0001
Mean (SD)	-6·0 (4·9)	-2·8 (3·9)		
[Range]	[-16 to 5]	[-12 to 4]		
Missing	*9*	*9*		
QIDS-SR$_{16}$ change at week 9				
Adjusted mean difference			-2·88 [-4·36 to -1·40]	0·00014
Mean (SD)	-5·9 (4·9)	-2·9 (4·3)		
[Range]	[-18 to 4]	[-15 to 6]		
Missing	*10*	*14*		
QIDS-SR$_{16}$ change at week 10				
Adjusted mean difference			-3·12 [-4·59 to -1·65]	<0·0001
Mean (SD)	-5·7 (4·7)	-2·7 (4·3)		
[Range]	[-18 to 4]	[-14 to 5]		
Missing	*8*	*13*		
QIDS-SR$_{16}$ change at week 11				
Adjusted mean difference			-2·92 [-4·41 to -1·43]	0·00012
Mean (SD)	-5·8 (5·1)	-2·8 (4·5)		
[Range]	[-18 to 6]	[-11 to 7]		
Missing	*14*	*13*		
QIDS-SR$_{16}$ change at week 16				
Adjusted mean difference			-3·47 [-5·00 to -1·95]	<0·0001
Mean (SD)	-6·2 (5·5)	-2·5 (4·3)		
[Range]	[-19 to 4]	[-15 to 7]		
Missing	*17*	*22*		
QIDS-SR$_{16}$ change at week 20				
Adjusted mean difference			-2·18 [-3·74 to -0·61]	0·0063
Mean (SD)	-6·1 (6·0)	-3·7 (4·6)		
[Range]	[-19 to 9]	[-15 to 6]		
Missing	*23*	*26*		

	Pramipexole (N=75)	Placebo (N=75)	Treatment effect [95% CI]*	P-value†
QIDS-SR$_{16}$ change at week 24				
Adjusted mean difference			-1·88 [-3·48 to -0·28]	0·021
Mean (SD)	-5·7 (5·7)	-3·3 (4·7)		
[Range]	[-17 to 9]	[-15 to 6]		
Missing	*25*	*32*		
QIDS-SR$_{16}$ change at week 28				
Adjusted mean difference			-2·77 [-4·37 to -1·18]	0·00067
Mean (SD)	-6·0 (6·0)	-3·2 (5·3)		
[Range]	[-18 to 4]	[-16 to 8]		
Missing	*24*	*32*		
QIDS-SR$_{16}$ change at week 32				
Adjusted mean difference			-3·19 [-4·82 to -1·55]	0·00013
Mean (SD)	-6·3 (5·6)	-2·9 (5·3)		
[Range]	[-19 to 10]	[-15 to 8]		
Missing	*30*	*34*		
QIDS-SR$_{16}$ change at week 36				
Adjusted mean difference			-2·75 [-4·42 to -1·08]	0·0012
Mean (SD)	-6·8 (5·5)	-4·0 (5·2)		
[Range]	[-18 to 5]	[-15 to 7]		
Missing	*32*	*38*		
QIDS-SR$_{16}$ change at week 40				
Adjusted mean difference			-2·25 [-3·94 to -0·55]	0·0093
Mean (SD)	-6·1 (5·8)	-3·8 (4·9)		
[Range]	[-16 to 9]	[-16 to 4]		
Missing	*34*	*40*		
QIDS-SR$_{16}$ change at week 44				
Adjusted mean difference			-2·33 [-4·09 to -0·56]	0·0098
Mean (SD)	-5·6 (5·7)	-3·3 (5·4)		
[Range]	[-16 to 4]	[-15 to 6]		
Missing	*44*	*40*		
QIDS-SR$_{16}$ change at week 48				
Adjusted mean difference			-2·02 [-3·73 to -0·31]	0·021
Mean (SD)	-6·1 (6·0)	-4·0 (5·1)		
[Range]	[-17 to 4]	[-15 to 3]		
Missing	*39*	*38*		
Discontinuation of trial treatment due to intolerance§				

	Pramipexole (N=75)	Placebo (N=75)	Treatment effect [95% CI]*	P-value†
Adjusted marginal relative risk			3·95 [1·35 to 11·57]	0·012
Yes, n(%)	15 (20·0%)	4 (5·3%)		
No, n(%)	60 (80·0%)	71 (94·7%)		
Missing	*0*	*0*		
Number of adverse reactions (ARs)‖				
Adjusted incidence rate ratio			2·24 [1·75 to 2·85]	<0·0001
Median (IQR)	6 (3 to 10)	2 (0 to 4)		
[Range]	[0 to 21]	[0 to 13]		
Missing	*0*	*0*		
Treatment satisfaction questionnaire for medication (TSQM-9) – Global satisfaction¶				
TSQM-9 global satisfaction at week 1				
Mean (SD)	51·7 (20·5)	43·9 (23·5)		
[Range]	[0·0 to 100·0]	[0·0 to 100·0]		
Missing	*33*	*35*		
TSQM-9 global satisfaction at week 6				
Mean (SD)	65·9 (28·1)	42·1 (27·6)		
[Range]	[7·1 to 100·0]	[0·0 to 100·0]		
Missing	*7*	*5*		
TSQM-9 global satisfaction at week 12				
Mean (SD)	65·6 (26·8)	35·4 (30·6)		
[Range]	[0·0 to 100·0]	[0·0 to 100·0]		
Missing	*10*	*11*		
TSQM-9 global satisfaction at week 24				
Mean (SD)	62·9 (31·1)	43·2 (31·6)		
[Range]	[0·0 to 100·0]	[0·0 to 100·0]		
Missing	*27*	*30*		
TSQM-9 global satisfaction at week 48				
Adjusted mean difference			26·41 [14·94 to 37·88]	<0·0001
Mean (SD)	69·4 (28·4)	38·3 (34·8)		
[Range]	[14·3 to 100·0]	[0·0 to 100·0]		
Missing	*43*	*45*		
Treatment satisfaction questionnaire for medication (TSQM-9) - Effectiveness¶				
TSQM-9 effectiveness at week 1				
Mean (SD)	45·6 (13·5)	39·3 (17·7)		
[Range]	[0·0 to 66·7]	[0·0 to 83·3]		
Missing	*33*	*35*		

	Pramipexole (N=75)	Placebo (N=75)	Treatment effect [95% CI]*	P-value†
TSQM-9 effectiveness at week 6				
Mean (SD)	65·0 (23·3)	36·3 (26·6)		
[Range]	[0·0 to 100·0]	[0·0 to 100·0]		
Missing	*7*	*5*		
TSQM-9 effectiveness at week 12				
Mean (SD)	65·2 (24·4)	34·2 (28·3)		
[Range]	[0·0 to 100·0]	[0·0 to 100·0]		
Missing	*10*	*11*		
TSQM-9 effectiveness at week 24				
Mean (SD)	62·3 (27·1)	39·5 (29·0)		
[Range]	[0·0 to 100·0]	[0·0 to 100·0]		
Missing	*27*	*30*		
TSQM-9 effectiveness at week 48				
Adjusted mean difference			34·31 [24·11 to 44·50]	<0·0001
Mean (SD)	68·9 (25·0)	35·0 (34·6)		
[Range]	[22·2 to 100·0]	[0·0 to 100·0]		
Missing	*43*	*45*		
Treatment satisfaction questionnaire for medication (TSQM-9) - Convenience¶				
TSQM-9 convenience at week 1				
Mean (SD)	86·4 (14·1)	85·8 (15·8)		
[Range]	[55·6 to 100·0]	[33·3 to 100·0]		
Missing	*33*	*35*		
TSQM-9 convenience at week 6				
Mean (SD)	82·5 (18·0)	83·8 (14·9)		
[Range]	[33·3 to 100·0]	[38·9 to 100·0]		
Missing	*7*	*5*		
TSQM-9 convenience at week 12				
Mean (SD)	85·5 (15·9)	83·4 (15·3)		
[Range]	[33·3 to 100·0]	[38·9 to 100·0]		
Missing	*10*	*11*		
TSQM-9 convenience at week 24				
Mean (SD)	81·9 (19·1)	83·3 (15·6)		
[Range]	[0·0 to 100·0]	[33·3 to 100·0]		
Missing	*27*	*30*		

	Pramipexole (N=75)	Placebo (N=75)	Treatment effect [95% CI]*	P-value†
TSQM-9 convenience at week 48				
Adjusted mean difference			1·75 [-5·04 to 8·54]	0·61
Mean (SD)	88·0 (12·5)	81·9 (18·6)		
[Range]	[55·6 to 100·0]	[33·3 to 100·0]		
Missing	*43*	*45*		
Altman self-rated scale for mania (ASRM)‡				
ASRM at baseline				
Mean (SD)	0·8 (1·3)	0·5 (1·1)		
[Range]	[0 to 8]	[0 to 4]		
Missing	*0*	*0*		
ASRM at week 1				
Adjusted mean difference			0·39 [-0·31 to 1·09]	0·28
Mean (SD)	1·1 (1·8)	0·6 (1·4)		
[Range]	[0 to 9]	[0 to 8]		
Missing	*3*	*2*		
ASRM at week 2				
Adjusted mean difference			0·87 [0·15 to 1·58]	0·017
Mean (SD)	1·8 (2·6)	0·7 (1·4)		
[Range]	[0 to 13]	[0 to 7]		
Missing	*7*	*7*		
ASRM at week 3				
Adjusted mean difference			0·94 [0·22 to 1·65]	0·010
Mean (SD)	2·3 (2·8)	1·2 (1·8)		
[Range]	[0 to 12]	[0 to 10]		
Missing	*6*	*8*		
ASRM at week 4				
Adjusted mean difference			1·76 [1·05 to 2·48]	<0·0001
Mean (SD)	2·8 (3·0)	0·8 (1·6)		
[Range]	[0 to 14]	[0 to 8]		
Missing	*6*	*8*		
ASRM at week 5				
Adjusted mean difference			2·05 [1·33 to 2·76]	<0·0001
Mean (SD)	3·0 (3·0)	0·8 (1·5)		
[Range]	[0 to 12]	[0 to 9]		
Missing	*7*	*7*		
ASRM at week 6				
Adjusted mean difference			1·58 [0·85 to 2·31]	<0·0001
Mean (SD)	2·7 (2·5)	1·0 (2·1)		
[Range]	[0 to 10]	[0 to 11]		
Missing	*12*	*9*		
ASRM at week 7				

	Pramipexole (N=75)	Placebo (N=75)	Treatment effect [95% CI]*	P-value†
Adjusted mean difference			1·25 [0·52 to 1·98]	0·00077
Mean (SD)	2·3 (2·8)	0·9 (1·7)		
[Range]	[0 to 10]	[0 to 10]		
Missing	*11*	*12*		
ASRM at week 8				
Adjusted mean difference			1·38 [0·65 to 2·11]	0·00020
Mean (SD)	2·2 (2·6)	0·7 (1·7)		
[Range]	[0 to 13]	[0 to 11]		
Missing	*9*	*12*		
ASRM at week 9				
Adjusted mean difference			1·23 [0·51 to 1·95]	0·00086
Mean (SD)	2·3 (2·6)	0·8 (2·0)		
[Range]	[0 to 10]	[0 to 11]		
Missing	*8*	*11*		
ASRM at week 10				
Adjusted mean difference			0·91 [0·18 to 1·64]	0·015
Mean (SD)	1·9 (2·2)	0·8 (1·8)		
[Range]	[0 to 10]	[0 to 10]		
Missing	*11*	*12*		
ASRM at week 11				
Adjusted mean difference			1·22 [0·49 to 1·95]	0·0011
Mean (SD)	2·3 (2·8)	1·0 (2·0)		
[Range]	[0 to 12]	[0 to 13]		
Missing	*12*	*12*		
ASRM at week 12				
Adjusted mean difference			1·09 [0·36 to 1·82]	0·0033
Mean (SD)	2·2 (2·5)	1·0 (2·3)		
[Range]	[0 to 10]	[0 to 11]		
Missing	*12*	*10*		
ASRM at week 16				
Adjusted mean difference			1·24 [0·48 to 2·00]	0·0014
Mean (SD)	2·2 (2·8)	0·7 (1·3)		
[Range]	[0 to 12]	[0 to 6]		
Missing	*17*	*20*		
ASRM at week 20				
Adjusted mean difference			0·41 [-0·38 to 1·19]	0·31
Mean (SD)	1·9 (2·3)	1·2 (2·2)		
[Range]	[0 to 10]	[0 to 11]		
Missing	*21*	*27*		
ASRM at week 24				
Adjusted mean difference			0·05 [-0·76 to 0·87]	0·89
Mean (SD)	1·8 (2·1)	1·3 (2·1)		

	Pramipexole (N=75)	Placebo (N=75)	Treatment effect [95% CI]*	P-value†
[Range]	[0 to 9]	[0 to 8]		
Missing	*26*	*32*		
ASRM at week 28				
Adjusted mean difference			0·50 [-0·31 to 1·30]	0·22
Mean (SD)	1·8 (2·3)	1·1 (1·8)		
[Range]	[0 to 10]	[0 to 6]		
Missing	*25*	*31*		
ASRM at week 32				
Adjusted mean difference			0·75 [-0·08 to 1·59]	0·078
Mean (SD)	2·3 (2·9)	1·3 (2·2)		
[Range]	[0 to 12]	[0 to 11]		
Missing	*29*	*36*		
ASRM at week 36				
Adjusted mean difference			1·01 [0·16 to 1·86]	0·020
Mean (SD)	2·5 (3·8)	1·2 (2·1)		
[Range]	[0 to 20]	[0 to 8]		
Missing	*31*	*38*		
ASRM at week 40				
Adjusted mean difference			0·91 [0·03 to 1·78]	0·042
Mean (SD)	2·0 (2·5)	0·8 (1·4)		
[Range]	[0 to 10]	[0 to 6]		
Missing	*37*	*39*		
ASRM at week 44				
Adjusted mean difference			1·33 [0·45 to 2·21]	0·0030
Mean (SD)	2·0 (3·0)	0·7 (1·1)		
[Range]	[0 to 9]	[0 to 4]		
Missing	*40*	*37*		
ASRM at week 48				
Adjusted mean difference			0·91 [-0·01 to 1·84]	0·054
Mean (SD)	2·1 (3·0)	1·1 (2·1)		
[Range]	[0 to 10]	[0 to 9]		
Missing	*42*	*45*		
Questionnaire for impulsive-compulsive disorder in Parkinson's disease-rating scale (QUIP-RS)‡				
QUIP-RS at baseline				
Mean (SD)	8·3 (8·2)	7·8 (9·5)		
[Range]	[0 to 43]	[0 to 40]		
Missing	*0*	*0*		
QUIP-RS at week 1				
Adjusted mean difference			0·64 [-1·26 to 2·55]	0·51
Mean (SD)	7·5 (8·6)	6·4 (9·0)		
[Range]	[0 to 37]	[0 to 33]		
Missing	*2*	*3*		
QUIP-RS at week 2				

	Pramipexole (N=75)	Placebo (N=75)	Treatment effect [95% CI]*	P-value†
Adjusted mean difference			0·66 [-1·25 to 2·58]	0·50
Mean (SD)	5·4 (7·7)	4·4 (7·3)		
[Range]	[0 to 37]	[0 to 32]		
Missing	*2*	*6*		
QUIP-RS at week 3				
Adjusted mean difference			-0·16 [-2·07 to 1·75]	0·87
Mean (SD)	5·0 (7·6)	5·1 (8·0)		
[Range]	[0 to 45]	[0 to 31]		
Missing	*3*	*3*		
QUIP-RS at week 4				
Adjusted mean difference			-0·21 [-2·13 to 1·71]	0·83
Mean (SD)	5·0 (7·7)	5·2 (8·2)		
[Range]	[0 to 45]	[0 to 30]		
Missing	*4*	*6*		
QUIP-RS at week 5				
Adjusted mean difference			-0·03 [-1·94 to 1·88]	0·97
Mean (SD)	4·6 (7·8)	4·5 (7·5)		
[Range]	[0 to 45]	[0 to 29]		
Missing	*4*	*4*		
QUIP-RS at week 6				
Adjusted mean difference			0·49 [-1·44 to 2·41]	0·62
Mean (SD)	4·9 (8·7)	4·0 (7·6)		
[Range]	[0 to 50]	[0 to 30]		
Missing	*6*	*6*		
QUIP-RS at week 7				
Adjusted mean difference			-0·16 [-2·10 to 1·77]	0·87
Mean (SD)	3·7 (6·1)	4·5 (8·5)		
[Range]	[0 to 22]	[0 to 40]		
Missing	*8*	*8*		
QUIP-RS at week 8				
Adjusted mean difference			0·19 [-1·74 to 2·12]	0·85
Mean (SD)	4·3 (6·7)	4·7 (8·9)		
[Range]	[0 to 25]	[0 to 36]		
Missing	*8*	*8*		
QUIP-RS at week 9				
Adjusted mean difference			-0·23 [-2·18 to 1·71]	0·81
Mean (SD)	3·8 (5·8)	4·3 (8·4)		
[Range]	[0 to 22]	[0 to 36]		
Missing	*9*	*11*		
QUIP-RS at week 10				
Adjusted mean difference			0·38 [-1·56 to 2·33]	0·70
Mean (SD)	3·6 (6·0)	4·0 (8·2)		

	Pramipexole (N=75)	Placebo (N=75)	Treatment effect [95% CI]*	P-value†
[Range]	[0 to 26]	[0 to 34]		
Missing	*10*	*11*		
QUIP-RS at week 11				
Adjusted mean difference			0·89 [-1·08 to 2·85]	0·38
Mean (SD)	3·8 (6·2)	3·4 (7·3)		
[Range]	[0 to 25]	[0 to 29]		
Missing	*16*	*11*		
QUIP-RS at week 12				
Adjusted mean difference			0·99 [-0·96 to 2·93]	0·32
Mean (SD)	3·8 (5·6)	3·6 (6·8)		
[Range]	[0 to 21]	[0 to 28]		
Missing	*8*	*11*		
QUIP-RS at week 16				
Adjusted mean difference			0·98 [-1·01 to 2·98]	0·33
Mean (SD)	4·9 (6·6)	4·1 (8·0)		
[Range]	[0 to 21]	[0 to 33]		
Missing	*19*	*20*		
QUIP-RS at week 20				
Adjusted mean difference			1·54 [-0·48 to 3·56]	0·14
Mean (SD)	4·9 (6·7)	3·9 (7·6)		
[Range]	[0 to 22]	[0 to 33]		
Missing	*20*	*25*		
QUIP-RS at week 24				
Adjusted mean difference			2·10 [0·06 to 4·13]	0·043
Mean (SD)	5·2 (6·8)	3·7 (7·4)		
[Range]	[0 to 28]	[0 to 28]		
Missing	*25*	*24*		
QUIP-RS at week 28				
Adjusted mean difference			1·77 [-0·32 to 3·85]	0·096
Mean (SD)	4·4 (6·8)	4·0 (8·1)		
[Range]	[0 to 26]	[0 to 32]		
Missing	*29*	*31*		
QUIP-RS at week 32				
Adjusted mean difference			1·37 [-0·71 to 3·46]	0·20
Mean (SD)	4·4 (6·6)	4·5 (8·4)		
[Range]	[0 to 19]	[0 to 32]		
Missing	*30*	*31*		
QUIP-RS at week 36				
Adjusted mean difference			1·06 [-1·09 to 3·20]	0·33
Mean (SD)	4·6 (6·1)	4·3 (8·7)		
[Range]	[0 to 19]	[0 to 29]		
Missing	*35*	*37*		
QUIP-RS at week 40				

	Pramipexole (N=75)	Placebo (N=75)	Treatment effect [95% CI]*	P-value†
Adjusted mean difference			0·63 [-1·56 to 2·82]	0·57
Mean (SD)	3·4 (6·1)	3·6 (6·9)		
[Range]	[0 to 24]	[0 to 23]		
Missing	*41*	*38*		
QUIP-RS at week 44				
Adjusted mean difference			1·03 [-1·15 to 3·22]	0·35
Mean (SD)	3·1 (5·4)	3·0 (6·2)		
[Range]	[0 to 21]	[0 to 21]		
Missing	*40*	*38*		
QUIP-RS at week 48				
Adjusted mean difference			0·91 [-1·26 to 3·09]	0·41
Mean (SD)	3·6 (6·0)	3·5 (7·1)		
[Range]	[0 to 22]	[0 to 26]		
Missing	*39*	*38*		
Suicidal ideation (QIDS-RS$_{16}$)‡				
Suicidal ideation at baseline				
Mean (SD)	1·0 (0·8)	1·1 (0·9)		
[Range]	[0 to 3]	[0 to 3]		
Missing	*0*	*0*		
Suicidal ideation at week 1				
Adjusted mean difference			-0·05 [-0·25 to 0·16]	0·66
Mean (SD)	0·9 (0·8)	0·9 (0·9)		
[Range]	[0 to 3]	[0 to 3]		
Missing	*6*	*6*		
Suicidal ideation at week 2				
Adjusted mean difference			-0·20 [-0·41 to 0·00]	0·052
Mean (SD)	0·7 (0·8)	0·9 (0·9)		
[Range]	[0 to 3]	[0 to 3]		
Missing	*7*	*6*		
Suicidal ideation at week 3				
Adjusted mean difference			-0·32 [-0·53 to -0·11]	0·0022
Mean (SD)	0·6 (0·7)	0·9 (0·8)		
[Range]	[0 to 2]	[0 to 3]		
Missing	*6*	*7*		
Suicidal ideation at week 4				
Adjusted mean difference			-0·45 [-0·65 to -0·25]	<0·0001
Mean (SD)	0·5 (0·6)	1·0 (1·0)		
[Range]	[0 to 2]	[0 to 3]		
Missing	*3*	*7*		

	Pramipexole (N=75)	Placebo (N=75)	Treatment effect [95% CI]*	P-value†
Suicidal ideation at week 5				
Adjusted mean difference			-0·49 [-0·69 to -0·28]	<0·0001
Mean (SD)	0·4 (0·6)	0·8 (0·9)		
[Range]	[0 to 2]	[0 to 3]		
Missing	*8*	*8*		
Suicidal ideation at week 6				
Adjusted mean difference			-0·45 [-0·66 to -0·24]	<0·0001
Mean (SD)	0·4 (0·7)	0·8 (0·9)		
[Range]	[0 to 2]	[0 to 3]		
Missing	*17*	*13*		
Suicidal ideation at week 7				
Adjusted mean difference			-0·51 [-0·72 to -0·30]	<0·0001
Mean (SD)	0·4 (0·6)	0·9 (1·0)		
[Range]	[0 to 2]	[0 to 3]		
Missing	*11*	*12*		
Suicidal ideation at week 8				
Adjusted mean difference			-0·38 [-0·59 to -0·17]	0·00036
Mean (SD)	0·4 (0·7)	0·8 (0·9)		
[Range]	[0 to 2]	[0 to 3]		
Missing	*9*	*9*		
Suicidal ideation at week 9				
Adjusted mean difference			-0·40 [-0·61 to -0·19]	0·00020
Mean (SD)	0·5 (0·7)	0·9 (0·9)		
[Range]	[0 to 2]	[0 to 3]		
Missing	*10*	*14*		
Suicidal ideation at week 10				
Adjusted mean difference			-0·53 [-0·74 to -0·32]	<0·0001
Mean (SD)	0·4 (0·6)	0·9 (0·9)		
[Range]	[0 to 2]	[0 to 3]		
Missing	*8*	*13*		
Suicidal ideation at week 11				
Adjusted mean difference			-0·44 [-0·65 to -0·23]	<0·0001
Mean (SD)	0·5 (0·7)	0·9 (1·0)		
[Range]	[0 to 3]	[0 to 3]		
Missing	*14*	*13*		
Suicidal ideation at week 12				

	Pramipexole (N=75)	Placebo (N=75)	Treatment effect [95% CI]*	P-value†
Adjusted mean difference			-0·48 [-0·68 to -0·27]	<0·0001
Mean (SD)	0·4 (0·7)	0·9 (0·9)		
[Range]	[0 to 3]	[0 to 3]		
Missing	*7*	*8*		
Suicidal ideation at week 16				
Adjusted mean difference			-0·41 [-0·63 to -0·19]	0·00023
Mean (SD)	0·4 (0·7)	0·8 (0·8)		
[Range]	[0 to 2]	[0 to 3]		
Missing	*17*	*22*		
Suicidal ideation at week 20				
Adjusted mean difference			-0·44 [-0·67 to -0·22]	0·00011
Mean (SD)	0·4 (0·7)	0·8 (0·9)		
[Range]	[0 to 2]	[0 to 3]		
Missing	*23*	*26*		
Suicidal ideation at week 24				
Adjusted mean difference			-0·37 [-0·60 to -0·14]	0·0014
Mean (SD)	0·4 (0·6)	0·9 (0·9)		
[Range]	[0 to 2]	[0 to 3]		
Missing	*25*	*32*		
Suicidal ideation at week 28				
Adjusted mean difference			-0·32 [-0·55 to -0·09]	0·0069
Mean (SD)	0·5 (0·8)	0·8 (0·9)		
[Range]	[0 to 2]	[0 to 3]		
Missing	*24*	*32*		
Suicidal ideation at week 32				
Adjusted mean difference			-0·28 [-0·52 to -0·05]	0·018
Mean (SD)	0·4 (0·7)	0·7 (0·9)		
[Range]	[0 to 3]	[0 to 3]		
Missing	*30*	*34*		
Suicidal ideation at week 36				
Adjusted mean difference			-0·49 [-0·74 to -0·25]	<0·0001
Mean (SD)	0·4 (0·6)	0·8 (0·9)		
[Range]	[0 to 3]	[0 to 3]		
Missing	*32*	*38*		
Suicidal ideation at week 40				
Adjusted mean difference			-0·36 [-0·60 to -0·11]	0·0041

	Pramipexole (N=75)	Placebo (N=75)	Treatment effect [95% CI]*	P-value†
Mean (SD)	0·5 (0·8)	0·7 (0·9)		
[Range]	[0 to 3]	[0 to 3]		
Missing	*34*	*40*		
Suicidal ideation at week 44				
Adjusted mean difference			-0·33 [-0·59 to -0·08]	0·011
Mean (SD)	0·5 (0·8)	0·8 (0·9)		
[Range]	[0 to 3]	[0 to 3]		
Missing	*44*	*40*		
Suicidal ideation at week 48				
Adjusted mean difference			-0·44 [-0·68 to -0·19]	0·00057
Mean (SD)	0·5 (0·8)	0·8 (1·0)		
[Range]	[0 to 3]	[0 to 3]		
Missing	*39*	*38*		
[Range]	[-6·9 to 6·2]	[-6·6 to 5·9]		
Missing	*10*	*12*		
QIDS-SR$_{16}$ response rate (reduction of ≥50% in baseline score) at week 12††				
Adjusted marginal relative risk			2·72 [1·49 to 4·95]	0·0011
Yes, n(%)	30 (44·1%)	11 (16·4%)		
No, n(%)	38 (55·9%)	56 (83·6%)		
Missing	*7*	*8*		
QIDS-SR$_{16}$ remission rate (score of ≤5) at week 12††				
Adjusted marginal relative risk			3·94 [1·61 to 9·64]	0·0026
Yes, n(%)	19 (27·9%)	5 (7·5%)		
No, n(%)	49 (72·1%)	62 (92·5%)		
Missing	*7*	*8*		
Anhedonia (SHAPS)‡				
SHAPS change at week 2				
Adjusted mean difference			-1·48 [-2·66 to -0·29]	0·014
Mean (SD)	-1·7 (3·4)	0·1 (2·6)		
[Range]	[-14 to 3]	[-6 to 9]		
Missing	*6*	*4*		
SHAPS change at week 4				
Adjusted mean difference			-1·79 [-2·98 to -0·60]	0·0031
Mean (SD)	-2·7 (4·1)	-0·8 (2·9)		
[Range]	[-14 to 8]	[-8 to 8]		
Missing	*4*	*7*		
SHAPS change at week 6				
Adjusted mean difference			-1·52 [-2·72 to -0·32]	0·013
Mean (SD)	-3·0 (3·7)	-1·5 (3·1)		

	Pramipexole (N=75)	Placebo (N=75)	Treatment effect [95% CI]*	P-value†
[Range]	[-14 to 7]	[-12 to 6]		
Missing	*10*	*7*		
SHAPS change at week 8				
Adjusted mean difference			-2·34 [-3·55 to -1·13]	0·00015
Mean (SD)	-3·0 (4·1)	-0·5 (3·7)		
[Range]	[-14 to 8]	[-9 to 12]		
Missing	*9*	*13*		
SHAPS change at week 10				
Adjusted mean difference			-2·87 [-4·08 to -1·66]	<0·0001
Mean (SD)	-3·4 (4·1)	-0·5 (4·0)		
[Range]	[-14 to 5]	[-8 to 12]		
Missing	*11*	*11*		
SHAPS change at week 12				
Adjusted mean difference			-2·20 [-3·41 to -1·00]	0·00035
Mean (SD)	-3·8 (4·0)	-1·5 (3·6)		
[Range]	[-14 to 3]	[-12 to 8]		
Missing	*10*	*10*		
SHAPS change at week 16				
Adjusted mean difference			-2·74 [-4·01 to -1·48]	<0·0001
Mean (SD)	-3·8 (4·3)	-0·7 (4·4)		
[Range]	[-14 to 5]	[-12 to 12]		
Missing	*17*	*24*		
SHAPS change at week 20				
Adjusted mean difference			-1·98 [-3·26 to -0·69]	0·0025
Mean (SD)	-4·1 (4·6)	-2·0 (4·4)		
[Range]	[-14 to 6]	[-12 to 8]		
Missing	*23*	*25*		
SHAPS change at week 24				
Adjusted mean difference			-1·22 [-2·53 to 0·10]	0·070
Mean (SD)	-4·0 (4·4)	-2·3 (4·6)		
[Range]	[-14 to 6]	[-12 to 10]		
Missing	*24*	*32*		
SHAPS change at week 28				
Adjusted mean difference			-1·67 [-2·99 to -0·35]	0·013
Mean (SD)	-3·6 (4·5)	-1·8 (4·1)		
[Range]	[-14 to 7]	[-9 to 7]		
Missing	*24*	*33*		

	Pramipexole (N=75)	Placebo (N=75)	Treatment effect [95% CI]*	P-value†
SHAPS change at week 32				
Adjusted mean difference			-2·98 [-4·32 to -1·63]	<0·0001
Mean (SD)	-4·3 (4·5)	-1·1 (4·3)		
[Range]	[-14 to 3]	[-9 to 7]		
Missing	*30*	*34*		
SHAPS change at week 36				
Adjusted mean difference			-1·93 [-3·31 to -0·56]	0·0059
Mean (SD)	-4·9 (4·1)	-2·9 (4·8)		
[Range]	[-14 to 3]	[-12 to 7]		
Missing	*32*	*38*		
SHAPS change at week 40				
Adjusted mean difference			-1·41 [-2·81 to 0·00]	0·050
Mean (SD)	-4·4 (4·2)	-2·7 (4·5)		
[Range]	[-14 to 6]	[-12 to 5]		
Missing	*35*	*41*		
SHAPS change at week 44				
Adjusted mean difference			-1·86 [-3·29 to -0·42]	0·011
Mean (SD)	-4·4 (3·7)	-1·9 (5·1)		
[Range]	[-14 to 1]	[-12 to 11]		
Missing	*42*	*39*		
SHAPS change at week 48				
Adjusted mean difference			-1·67 [-3·10 to -0·25]	0·022
Mean (SD)	-4·3 (3·9)	-2·6 (4·5)		
[Range]	[-14 to 1]	[-12 to 6]		
Missing	*40*	*40*		
Anxiety (GAD-7)‡				
GAD-7 change at week 1				
Adjusted mean difference			-1·14 [-2·55 to 0·27]	0·11
Mean (SD)	-1·2 (3·8)	-0·3 (2·6)		
[Range]	[-11 to 10]	[-10 to 6]		
Missing	*3*	*2*		
GAD-7 change at week 2				
Adjusted mean difference			-1·09 [-2·53 to 0·34]	0·14
Mean (SD)	-2·1 (3·6)	-0·8 (3·1)		
[Range]	[-14 to 4]	[-8 to 7]		
Missing	*7*	*7*		
GAD-7 change at week 3				
Adjusted mean difference			-1·55 [-2·99 to -0·12]	0·034

	Pramipexole (N=75)	Placebo (N=75)	Treatment effect [95% CI]*	P-value†
Mean (SD) [Range] *Missing*	-2·8 (4·4) [-16 to 10] *6*	-1·3 (3·8) [-10 to 7] *8*		
GAD-7 change at week 4				
Adjusted mean difference			-1·90 [-3·34 to -0·46]	0·0095
Mean (SD) [Range] *Missing*	-3·3 (4·8) [-16 to 7] *7*	-1·3 (4·1) [-12 to 10] *8*		
GAD-7 change at week 5				
Adjusted mean difference			-1·90 [-3·33 to -0·46]	0·0095
Mean (SD) [Range] *Missing*	-3·6 (4·8) [-18 to 9] *6*	-1·6 (4·2) [-13 to 12] *7*		
GAD-7 change at week 6				
Adjusted mean difference			-1·69 [-3·15 to -0·24]	0·022
Mean (SD) [Range] *Missing*	-3·2 (5·4) [-18 to 10] *12*	-1·7 (4·4) [-15 to 12] *9*		
GAD-7 change at week 7				
Adjusted mean difference			-1·13 [-2·59 to 0·33]	0·13
Mean (SD) [Range] *Missing*	-2·9 (5·0) [-18 to 8] *11*	-1·5 (4·7) [-15 to 12] *12*		
GAD-7 change at week 8				
Adjusted mean difference			-1·87 [-3·33 to -0·42]	0·012
Mean (SD) [Range] *Missing*	-3·2 (5·2) [-19 to 6] *9*	-1·2 (4·1) [-15 to 8] *12*		
GAD-7 change at week 9				
Adjusted mean difference			-2·24 [-3·69 to -0·80]	0·0024
Mean (SD) [Range] *Missing*	-4·0 (5·3) [-19 to 9] *8*	-1·7 (4·5) [-13 to 11] *11*		
GAD-7 change at week 10				
Adjusted mean difference			-2·31 [-3·77 to -0·85]	0·0019
Mean (SD) [Range] *Missing*	-4·1 (5·4) [-19 to 8] *11*	-1·7 (4·4) [-13 to 10] *12*		
GAD-7 change at week 11				
Adjusted mean difference			-2·47 [-3·93 to -1·01]	0·00093
Mean (SD)	-4·0 (5·5)	-1·4 (4·9)		

	Pramipexole (N=75)	Placebo (N=75)	Treatment effect [95% CI]*	P-value†
[Range]	[-19 to 6]	[-14 to 13]		
Missing	*12*	*12*		
GAD-7 change at week 12				
Adjusted mean difference			-2·51 [-3·97 to -1·06]	0·00073
Mean (SD)	-4·1 (5·5)	-1·5 (4·7)		
[Range]	[-19 to 7]	[-17 to 11]		
Missing	*12*	*10*		
GAD-7 change at week 16				
Adjusted mean difference			-2·99 [-4·49 to -1·49]	<0·0001
Mean (SD)	-4·5 (5·5)	-1·3 (4·6)		
[Range]	[-17 to 6]	[-17 to 12]		
Missing	*17*	*20*		
GAD-7 change at week 20				
Adjusted mean difference			-2·22 [-3·76 to -0·68]	0·0047
Mean (SD)	-4·1 (5·7)	-2·0 (5·3)		
[Range]	[-19 to 7]	[-17 to 12]		
Missing	*21*	*27*		
GAD-7 change at week 24				
Adjusted mean difference			-0·96 [-2·54 to 0·62]	0·23
Mean (SD)	-3·6 (5·9)	-2·7 (5·0)		
[Range]	[-19 to 11]	[-17 to 7]		
Missing	*26*	*32*		
GAD-7 change at week 28				
Adjusted mean difference			-1·60 [-3·17 to -0·03]	0·046
Mean (SD)	-4·6 (5·8)	-3·0 (4·5)		
[Range]	[-19 to 4]	[-17 to 3]		
Missing	*25*	*30*		
GAD-7 change at week 32				
Adjusted mean difference			-2·16 [-3·78 to -0·55]	0·0088
Mean (SD)	-4·5 (6·1)	-2·4 (5·1)		
[Range]	[-20 to 12]	[-17 to 7]		
Missing	*29*	*36*		
GAD-7 change at week 36				
Adjusted mean difference			-1·92 [-3·56 to -0·28]	0·022
Mean (SD)	-4·9 (5·8)	-3·5 (4·7)		
[Range]	[-20 to 5]	[-16 to 7]		
Missing	*31*	*38*		

	Pramipexole (N=75)	Placebo (N=75)	Treatment effect [95% CI]*	P-value†
GAD-7 change at week 40				
Adjusted mean difference			-2·20 [-3·88 to -0·52]	0·010
Mean (SD)	-4·6 (5·5)	-2·3 (5·1)		
[Range]	[-15 to 8]	[-17 to 11]		
Missing	*37*	*39*		
GAD-7 change at week 44				
Adjusted mean difference			-2·35 [-4·03 to -0·67]	0·0061
Mean (SD)	-4·4 (6·4)	-2·3 (4·7)		
[Range]	[-17 to 7]	[-17 to 9]		
Missing	*39*	*37*		
GAD-7 change at week 48				
Adjusted mean difference			-2·07 [-3·83 to -0·30]	0·022
Mean (SD)	-4·3 (6·1)	-3·4 (4·8)		
[Range]	[-17 to 8]	[-17 to 6]		
Missing	*42*	*45*		
Clinician rated depression (QIDS-C)‡				
QIDS-C change at week 2				
Adjusted mean difference			-1·40 [-2·76 to -0·03]	0·045
Mean (SD)	-2·9 (3·5)	-1·5 (3·4)		
[Range]	[-15 to 5]	[-13 to 6]		
Missing	*6*	*3*		
QIDS-C change at week 6				
Adjusted mean difference			-3·69 [-5·08 to -2·29]	<0·0001
Mean (SD)	-6·0 (4·1)	-2·3 (5·0)		
[Range]	[-14 to 4]	[-16 to 10]		
Missing	*9*	*9*		
QIDS-C change at week 12				
Adjusted mean difference			-3·26 [-4·67 to -1·86]	<0·0001
Mean (SD)	-6·1 (4·5)	-3·0 (4·9)		
[Range]	[-18 to 6]	[-16 to 5]		
Missing	*10*	*9*		
QIDS-C change at week 48				
Adjusted mean difference			-3·18 [-4·91 to -1·44]	0·00033
Mean (SD)	-7·0 (6·0)	-4·0 (6·4)		
[Range]	[-17 to 4]	[-17 to 7]		
Missing	*38*	*39*		
Work and social adjustment scale (WSAS)‡				

	Pramipexole (N=75)	Placebo (N=75)	Treatment effect [95% CI]*	P-value†
WSAS change at week 4				
Adjusted mean difference			-3·02 [-5·54 to -0·50]	0·019
Mean (SD)	-4·2 (9·3)	-1·0 (5·5)		
[Range]	[-28 to 18]	[-15 to 10]		
Missing	*4*	*7*		
WSAS change at week 8				
Adjusted mean difference			-4·78 [-7·36 to -2·20]	0·00028
Mean (SD)	-5·8 (9·3)	-0·3 (6·6)		
[Range]	[-32 to 12]	[-18 to 16]		
Missing	*10*	*12*		
WSAS change at week 12				
Adjusted mean difference			-4·45 [-7·01 to -1·88]	0·00067
Mean (SD)	-7·3 (8·7)	-2·3 (6·3)		
[Range]	[-33 to 7]	[-20 to 10]		
Missing	*9*	*10*		
WSAS change at week 24				
Adjusted mean difference			-3·44 [-6·19 to -0·69]	0·014
Mean (SD)	-6·5 (9·5)	-2·5 (6·9)		
[Range]	[-33 to 8]	[-22 to 16]		
Missing	*22*	*29*		
WSAS change at week 36				
Adjusted mean difference			-4·12 [-7·03 to -1·21]	0·0056
Mean (SD)	-8·0 (10·3)	-3·6 (6·5)		
[Range]	[-33 to 9]	[-18 to 9]		
Missing	*30*	*39*		
WSAS change at week 48				
Adjusted mean difference			-3·23 [-6·18 to -0·27]	0·032
Mean (SD)	-6·8 (11·4)	-3·3 (6·9)		
[Range]	[-34 to 12]	[-18 to 7]		
Missing	*37*	*37*		

**Pramipexole versus Placebo. †Level of statistical significance = 0·05. ‡Mixed effects linear regression model adjusted for randomised group, timepoint, age, sex, baseline QIDS-SR$_{16}$ score, an interaction between randomised group and timepoint, and baseline outcome score as fixed effects, site as a random effect, and participant ID as a random intercept. A negative mean difference indicates improvement in favour of Pramipexole. §Mixed effects logistic regression model adjusted for randomised group, age, sex, and baseline QIDS-SR$_{16}$ score as fixed effects, and site as a random effect. A marginal relative risk less than 1 indicates improvement in favour of Pramipexole. ‖Mixed effects Poisson regression model adjusted for randomised group, age, sex, and baseline QIDS-SR$_{16}$ score as fixed effects, and site as a random effect. An incidence rate ratio less than 1 indicates improvement in favour of Pramipexole. ¶Mixed effects linear regression model adjusted for randomised group, timepoint, age, sex, baseline QIDS-SR$_{16}$ score, and an interaction between randomised group and timepoint as fixed effects, site as a random effect, and participant ID as a random intercept. A positive mean difference indicates improvement in favour of Pramipexole. ††Mixed effects logistic regression model adjusted for*

randomised group, age, sex, and baseline QIDS-SR$_{16}$ score as fixed effects, and site as a random effect. A marginal relative risk greater than 1 indicates improvement in favour of Pramipexole.

Consort Checklist

CONSORT 2010 checklist of information to include when reporting a randomised trial*

Section/Topic	Item No	Checklist item	Reported on page No
Title and abstract			
	1a	Identification as a randomised trial in the title	1
	1b	Structured summary of trial design, methods, results, and conclusions (for specific guidance see CONSORT for abstracts)	3-4
Introduction			
Background and objectives	2a	Scientific background and explanation of rationale	7-8
	2b	Specific objectives or hypotheses	7-8
Methods			
Trial design	3a	Description of trial design (such as parallel, factorial) including allocation ratio	8
	3b	Important changes to methods after trial commencement (such as eligibility criteria), with reasons	8 and 12
Participants	4a	Eligibility criteria for participants	9
	4b	Settings and locations where the data were collected	8-9
Interventions	5	The interventions for each group with sufficient details to allow replication, including how and when they were actually administered	11

Outcomes	6a	Completely defined pre-specified primary and secondary outcome measures, including how and when they were assessed	12-13
	6b	Any changes to trial outcomes after the trial commenced, with reasons	12
Sample size	7a	How sample size was determined	13-14
	7b	When applicable, explanation of any interim analyses and stopping guidelines	NA
Randomisation:			
Sequence generation	8a	Method used to generate the random allocation sequence	10
	8b	Type of randomisation; details of any restriction (such as blocking and block size)	10
Allocation concealment mechanism	9	Mechanism used to implement the random allocation sequence (such as sequentially numbered containers), describing any steps taken to conceal the sequence until interventions were assigned	10-11
Implementation	10	Who generated the random allocation sequence, who enrolled participants, and who assigned participants to interventions	10
Blinding	11a	If done, who was blinded after assignment to interventions (for example, participants, care providers, those assessing outcomes) and how	10-11
	11b	If relevant, description of the similarity of interventions	11
Statistical methods	12a	Statistical methods used to compare groups for primary and secondary outcomes	13-14 and appendix
	12b	Methods for additional analyses, such as subgroup analyses and adjusted analyses	appendix

Results

Participant flow (a diagram is strongly recommended)	13a	For each group, the numbers of participants who were randomly assigned, received intended treatment, and were analysed for the primary outcome	14-16 (figure 1)
	13b	For each group, losses and exclusions after randomisation, together with reasons	14-16 (figure 1)
Recruitment	14a	Dates defining the periods of recruitment and follow-up	14
	14b	Why the trial ended or was stopped	14
Baseline data	15	A table showing baseline demographic and clinical characteristics for each group	18-19 (table 1)
Numbers analysed	16	For each group, number of participants (denominator) included in each analysis and whether the analysis was by original assigned groups	14-15 (figure 1)
Outcomes and estimation	17a	For each primary and secondary outcome, results for each group, and the estimated effect size and its precision (such as 95% confidence interval)	19-21 Table 2
	17b	For binary outcomes, presentation of both absolute and relative effect sizes is recommended	19 table 2
Ancillary analyses	18	Results of any other analyses performed, including subgroup analyses and adjusted analyses, distinguishing pre-specified from exploratory	appendix
Harms	19	All important harms or unintended effects in each group (for specific guidance see CONSORT for harms)	23-25, Table 3, appendix

Discussion

Limitations	20	Trial limitations, addressing sources of potential bias, imprecision, and, if relevant, multiplicity of analyses	28
Generalisability	21	Generalisability (external validity, applicability) of the trial findings	28-29
Interpretation	22	Interpretation consistent with results, balancing benefits and harms, and considering other relevant evidence	26-29
Other information			
Registration	23	Registration number and name of trial registry	8
Protocol	24	Where the full trial protocol can be accessed, if available	8 (reference number 13)
Funding	25	Sources of funding and other support (such as supply of drugs), role of funders	4, 32

Citation: Schulz KF, Altman DG, Moher D, for the CONSORT Group. CONSORT 2010 Statement: updated guidelines for reporting parallel group randomised trials. BMC Medicine. 2010;8:18.
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*We strongly recommend reading this statement in conjunction with the CONSORT 2010 Explanation and Elaboration for important clarifications on all the items. If relevant, we also recommend reading CONSORT extensions for cluster randomised trials, non-inferiority and equivalence trials, non-pharmacological treatments, herbal interventions, and pragmatic trials. Additional extensions are forthcoming: for those and for up-to-date references relevant to this checklist, see www.consort-statement.org.

Supplementary references

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